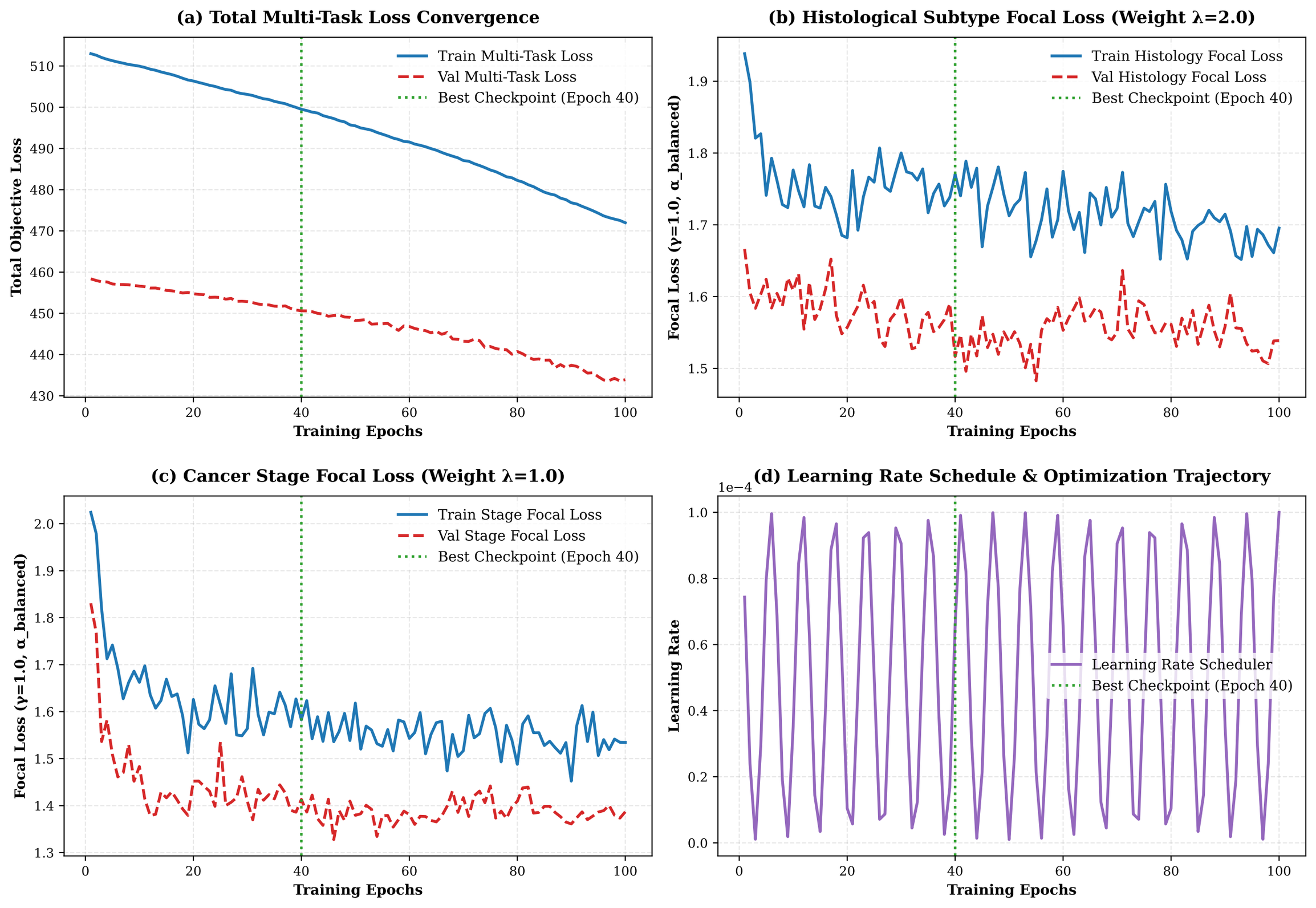
(a) Total Multi-Task Loss Convergence
Train Multi-Task Loss
Val Multi-Task Loss
Best Checkpoint (Epoch 40)
Total Objective Loss
Training Epochs
(b) Histological Subtype Focal Loss (Weight λ=2.0)
Train Histology Focal Loss
Val Histology Focal Loss
Best Checkpoint (Epoch 40)
Focal Loss (γ=1.0, α_balanced)
Training Epochs
(c) Cancer Stage Focal Loss (Weight λ=1.0)
Train Stage Focal Loss
Val Stage Focal Loss
Best Checkpoint (Epoch 40)
Focal Loss (γ=1.0, α_balanced)
Training Epochs
(d) Learning Rate Schedule & Optimization Trajectory
1e−4
Learning Rate Scheduler
Best Checkpoint (Epoch 40)
Learning Rate
Training Epochs